

Phase and Topology of Distributed CO₂ Dynamics in a Basalt Hillslope at the Landscape Evolution Observatory

Theoretical framework, mathematical definitions,
and a reproducible analysis protocol

Abstract

This document develops a framework for investigating the transition from crushed basalt to incipient soil using a distributed network of CO₂ sensors at the Landscape Evolution Observatory (LEO), Biosphere 2. The analysis takes as its starting point the observed delays between CO₂ signals in the air above the surface and at depths of approximately 5, 25, and 50 cm. A diffusion–reaction model provides the physical reference; cross-spectral or wavelet analysis yields relative phases; and a directed phase-transport graph describes relationships across the sensor network. For recurring rainfall–wetting–drainage–drying cycles, the framework defines a geometric phase along the trajectory of a normalized complex state. Persistent homology characterizes the spatial organization of regions with elevated or depleted CO₂ concentrations. A complementary component uses V. M. Buchstaber’s multidimensional time-series unfolding method: overlapping segments define an oriented curve in delay-coordinate space, scatter-matrix eigenvalues and a q -width quantify its effective dimensionality, and optimal projection subspaces are compared on a Grassmann manifold. Together, the phase, geometric, network, and topological descriptors form a phase–topology signature. A central objective is to distinguish reversible moisture responses from irreversible drift associated with weathering, secondary mineral formation, changes in pore connectivity, and biological colonization. The geometric-phase component draws on ideas developed by P. A. Deymier, while retaining the distinction between a scalar gas-concentration field and topological acoustics. The proposed contribution is a testable integration of phase analysis and unfolding methods for depth-resolved pore-gas CO₂ records; Buchstaber’s method itself is established theory.

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1 Purpose, scope, and limits

The aim is to identify and quantify changes in CO_2 transport, production, and consumption as a basalt substrate develops. Concentration is not a direct measure of soil formation. It reflects the combined effects of at least four processes, summarized schematically as

$$\begin{aligned} \text{observed } \text{CO}_2 &\leftarrow \text{biological production} - \text{geochemical consumption} \\ &\quad - \text{diffusive export} - \text{dissolution and aqueous transport.} \end{aligned} \quad (1)$$

This is a conceptual summary, not a dimensionally closed mass-balance equation; the governing balance is given in (13). Interpretation must therefore consider the spatial and temporal structure of the response alongside independent measurements of moisture, temperature, water potential, electrical resistivity, solution chemistry, and biological activity. The direction of change in mean concentration alone is insufficient.

The term *phase-topology* refers to two related aspects of the analysis:

- 1) *signal phase*: the position of an oscillatory or event-related component relative to signals at other depths;
- 2) *field topology*: connectivity, connected components, and cycles in the spatial CO_2 field or in the system's trajectory through state space.

Here, phase has no connection to thermodynamic phases or the Gibbs phase rule. Time lags, phase shifts, and topological invariants are also distinct quantities. The framework moves from lag estimation to phase relationships and then to geometric and topological descriptions.

2 Measurement geometry at LEO

Consider a hillslope with horizontal sensor positions $x_m, m = 1, \dots, M$. At each position, assume a vertical measurement column with levels

$$z_0 = \text{air above the surface}, \quad z_1 \approx 5 \text{ cm}, \quad z_2 \approx 25 \text{ cm}, \quad z_3 \approx 50 \text{ cm}. \quad (2)$$

The observed concentration is

$$C_{m\ell}(t) = C(x_m, z_\ell, t), \quad \ell \in \{0, 1, 2, 3\}. \quad (3)$$

For a single column, define

$$\mathbf{C}_m(t) = [C_{m0}(t) \ C_{m1}(t) \ C_{m2}(t) \ C_{m3}(t)]^\top. \quad (4)$$

Stacking the columns gives the network state at time t :

$$\mathbf{C}(t) = [\mathbf{C}_1^\top(t) \ \dots \ \mathbf{C}_M^\top(t)]^\top \in \mathbb{R}^N, \quad N = 4M \quad (5)$$

when all four levels are available. With incomplete coverage, N denotes the number of channels retained after quality control.

Early LEO pore-gas CO_2 measurements were collected every 15 minutes at 48 positions collocated with moisture sensors, temperature sensors, and solution samplers [7]. The configuration varied across experiments. Coordinates, depths, units, measurement ranges, and periods of availability must therefore be read from the metadata for the actual dataset.

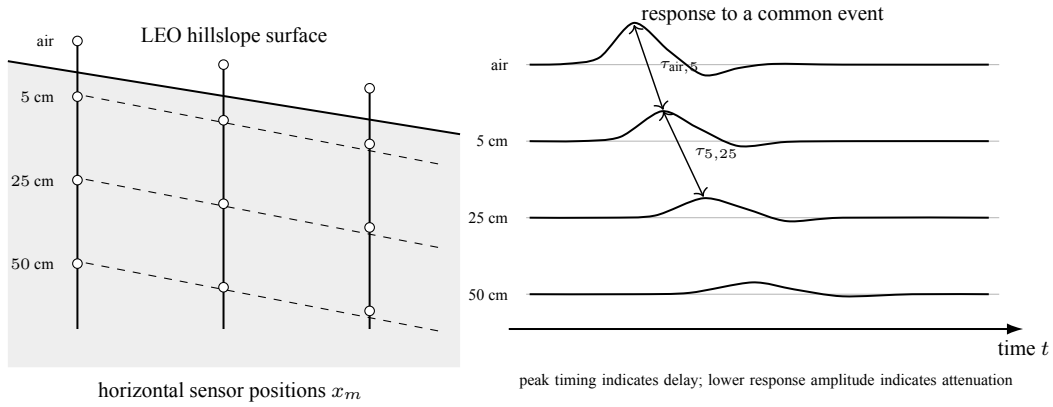


Figure 1: Measurement geometry and a schematic response. Left: vertical sensor columns at several hillslope positions. Right: an event response appears at successive levels with different delays and amplitudes. The diagram defines indices m and ℓ ; it does not show measured data.

3 Research questions and testable hypotheses

3.1 Research questions

- Q1. Are the vertical CO_2 delays consistent with effective diffusion, or do they require local sources, sinks, or advection?
- Q2. How do connectivity and transport direction between the air, shallow substrate, and deeper layers change after rainfall?
- Q3. Does the system return to its previous phase–topology configuration after drying?
- Q4. Can irreversible changes in that configuration be linked to mineralogical, hydrological, and biological development of the substrate?

3.2 Hypotheses

- H1. In a diffusion-dominated regime, phase delay increases monotonically with depth and is accompanied by consistent amplitude attenuation.
- H2. Preferential water and gas pathways produce unusually short lags, high coherence between widely separated depths, and spatial agreement with electrical resistivity tomography (ERT).
- H3. Local biological production can produce a nonmonotonic phase ordering, including a deep signal that leads the shallow signal.
- H4. After moisture and temperature have been matched, residual drift in topological descriptors reflects irreversible substrate reorganization.
- H5. A shift from predominantly abiotic CO_2 consumption to biological production is accompanied by a persistent reorganization of the phase-transport graph.

These hypotheses require independent validation: the proposed signal patterns do not uniquely identify their underlying mechanisms.

4 Mathematical preliminaries

The definitions below follow the order needed for the analysis: the phase of a single signal, relative phase between channels, a multichannel state, and finally its geometric and topological descriptors.

4.1 Time lag, phase shift, and topology

Time lag. The lag τ_{ij} is the delay of channel j relative to channel i , measured in units of time. If

$$x_j(t) = x_i(t - \tau), \quad \tau > 0, \quad (6)$$

then a change appears in i first and in j a time τ later.

Phase shift. For a harmonic component with angular frequency ω , the same delay gives

$$\Delta\phi = -\omega\tau \quad (7)$$

under the cross-spectrum convention $S_{ij} = X_i^* X_j$ used below. A phase shift is ambiguous without its frequency: the same $\Delta\phi$ represents different time lags at different frequencies.

Topological descriptor. Topology describes the organization of an object rather than the magnitude of a delay: its connected components, cycles, cavities, or the homotopy class of a trajectory. Such descriptors can be constructed from consistent phase relationships or spatial data, but they are not alternative names for time lags.

Example. For a daily period $T = 24$ h, a delay $\tau = 3$ h corresponds to $\Delta\phi = -2\pi\tau/T = -\pi/4$. This describes a pair of channels. By contrast, if a closed relative-phase trajectory makes one positive circuit around a coordinate of the phase torus during a rainfall cycle, its winding number in that coordinate is one. Closure and a well-defined continuous phase branch are required for this integer-valued interpretation.

4.2 Complex numbers, amplitude, and phase

The Cartesian and polar representations of a complex number are

$$z = x + iy = Ae^{i\phi}, \quad A = |z| = \sqrt{x^2 + y^2}, \quad \phi = \arg z = \text{atan2}(y, x). \quad (8)$$

Here $i^2 = -1$, x and y are the real and imaginary parts, A is the modulus, and ϕ is the angle measured from the positive real axis. Phase is defined only for $z \neq 0$. Euler's identity,

$$e^{i\phi} = \cos \phi + i \sin \phi, \quad (9)$$

allows amplitude and position within an oscillation to be encoded together.

Example. The numbers $z_1 = 2$ and $z_2 = 2i$ both have modulus 2, but their phases are 0 and $\pi/2$. Equal response amplitudes in two CO₂ channels therefore need not indicate the same stage of a dynamic cycle.

4.3 The argument and phase unwrapping

The function $\arg z$ usually returns a principal value in $(-\pi, \pi]$. A phase record may consequently contain artificial jumps of 2π . Phase unwrapping adds or subtracts multiples of 2π to obtain a continuous branch where the data support one.

Example. The principal values $170^\circ, 179^\circ, -175^\circ, -160^\circ$ unwrap to $170^\circ, 179^\circ, 185^\circ, 200^\circ$. Unwrapping must not bridge intervals with low amplitude or low coherence: the phase branch is not identifiable there. Sampling must also be sufficient to resolve changes between adjacent observations.

4.4 Bra–ket notation and the complex inner product

A *ket*, $|\psi\rangle$, denotes a complex state vector. Its corresponding *bra*, $\langle\psi|$, is the Hermitian transpose: transpose the vector and conjugate each component. For

$$|u\rangle = [u_1 \ \cdots \ u_N]^T, \quad |v\rangle = [v_1 \ \cdots \ v_N]^T, \quad (10)$$

the complex inner product and norm are

$$\langle u|v\rangle = \sum_{j=1}^N u_j^* v_j, \quad \|u\|_2 = \sqrt{\langle u|u\rangle}. \quad (11)$$

The asterisk denotes complex conjugation: $(a + ib)^* = a - ib$. Bra–ket notation is used solely as compact linear algebra; it assigns no quantum-mechanical properties to CO_2 .

Example. Let $|u\rangle = (1, i)^T/\sqrt{2}$ and $|v\rangle = (1, 1)^T/\sqrt{2}$. Then

$$\langle u|v\rangle = \frac{1-i}{2}, \quad |\langle u|v\rangle| = \frac{1}{\sqrt{2}}, \quad \arg\langle u|v\rangle = -\frac{\pi}{4}. \quad (12)$$

For normalized states, the modulus measures their overlap; the argument records their relative phase under the chosen representatives.

5 Rationale and proposed contribution

5.1 What concentrations and individual lags leave unresolved

Mean concentration and individual time lags are useful but incomplete descriptions of a distributed system. The same mean CO_2 concentration can result from different combinations of local production, geochemical consumption, dissolution, gas diffusion, and displacement by water. Likewise, a single pairwise lag does not establish the order in which an entire column responds or whether the system returns to its previous configuration after an event.

The proposed framework adds two levels of information. Relative phases describe the sequence of changes across the network. Trajectory geometry and topology describe the full cycle, including the order of intermediate states, loops, hysteresis, and residual drift. These descriptors complement concentration measurements, mass balances, and physical models within a multi-channel description.

Example. Two rainfall events may begin and end at the same concentrations. During one, the signal is progressively delayed from 5 to 25 to 50 cm. During the other, the deepest channel temporarily leads the middle channel because of a local source or preferential pathway. Endpoint concentrations cannot distinguish these histories; the relative-phase trajectory and phase-transport graph can.

5.2 Why use phase?

Phase expresses delay as a dimensionless circular coordinate, allowing coherent responses to be compared across different amplitudes. Removing a common phase leaves relative phases in a product of circles, naturally represented by the torus $\mathbb{T}^{N-1}$. This permits winding-number analysis. A related complex-state representation supports geometric-phase analysis. Geometric phase depends on the path through state space, not just its endpoints. Its gauge invariance follows from cancellation of arbitrary local phase factors in the closed product in (55) [1].

Phase is not universally robust to noise. It is useful only where amplitudes are adequate, coherence is statistically significant, and results persist across reasonable choices of window, filter, and sensor subset. Amplitude and coherence masks are therefore part of the definition of admissible phase data.

5.3 What topology contributes

Topological descriptors capture properties preserved under appropriate continuous deformations, such as connectivity, cycles, and winding classes. In this study, topology and the accompanying trajectory geometry offer three distinct benefits:

1. the number and persistence of spatial CO₂ domains can be compared as their shapes and amplitudes change;
2. trajectory closure and shape help distinguish reversible cycles, hysteresis, and residual drift;
3. admissible joint phase relationships can be represented by a simplicial complex, extending the description beyond pairwise correlations.

Persistent descriptors still depend on the filtration and its scale; they are not invariant under arbitrary changes in amplitude. Likewise, closure and hysteresis require geometric and physical interpretation, not topology alone.

The third benefit requires particular care. High pairwise coherence does not by itself establish joint consistency across a group of channels. A simplicial representation requires explicit compatibility criteria for pairs, triples, and larger groups, with every admissible simplex also containing all its faces.

5.4 The proposed methodological contribution

Novelty should be stated as a methodological proposal, not as an established physical finding. The proposed integration comprises:

1. adapting geometric-phase reasoning developed for wave and acoustic states to a distributed scalar gas-concentration field governed by an explicit diffusion–reaction model;
2. deriving phase coordinates from measured delays between specified physical depths;

3. combining amplitude attenuation, phase gradients, a directed transport graph, event geometric phases, and persistent topology of the spatial field;
4. comparing signatures at matched dry states to distinguish reversible hydrological responses from long-term substrate change;
5. incorporating Buchstaber’s multidimensional unfolding method to assess effective dimensionality, compare optimal subspaces on $G(n, q)$, and construct testable L -approximations of the records.

The final component uses the published theory of unfoldings [3, 4]. Its integration with phase lags in depth-resolved pore-gas CO₂ is proposed here. An absolute priority claim would require a systematic literature review; the appropriate claim is a new application and testable adaptation.

6 Notation

Symbol	Definition	Units/ domain
x_m	Horizontal position of a sensor column	m
z_ℓ	Measurement depth or height; the positive direction must be specified	m
t	Time	s or timestamp
$C_{m\ell}(t)$	Measured CO ₂ concentration	ppm or mol m ⁻³
$c_{m\ell}(t)$	Transformed, normalized concentration anomaly	1
$T_{m\ell}(t)$	Temperature	K or °C
$\theta_{m\ell}(t)$	Volumetric water content	m ³ m ⁻³
$\psi_{m\ell}(t)$	Water potential	Pa
D_g	Molecular diffusivity of CO ₂ in free air	m ² s ⁻¹
D_{eff}	Effective gas diffusivity in the porous medium, using the convention in (13)	m ² s ⁻¹
q_g	Advective gas flux per unit bulk cross-sectional area	m s ⁻¹
R	Net local source: production minus consumption	mol m ⁻³ s ⁻¹
$\omega = 2\pi f$	Angular frequency	rad s ⁻¹
A_j	Local amplitude of the selected temporal component	As input signal
ϕ_j	Local analytic-signal phase	rad
$\Delta\phi_{ij}$	Relative phase of channels i and j	rad mod 2π
τ_{ij}	Time lag between channels	s
γ_{ij}^2	Magnitude-squared coherence	[0, 1]
$z_j^{(a)}$	Complex analytic signal; superscript (a) distinguishes it from depth	$\mathbb{C}$
$ \psi(t)\rangle$	Normalized complex network-state vector	$\mathbb{C}^N$
η	Geometric phase of a closed cycle	rad mod 2π
$X_f(N; n)$	n -dimensional unfolding of a record of length N	Curve in $\mathbb{R}^n$

Symbol	Definition	Units/ domain
$G(n, q)$	Grassmann manifold of q -dimensional linear subspaces of $\mathbb{R}^n$	Manifold
$d_q(f; n)$	Normalized q -width for the quadratic unfolding objective	Squared units of f
$\hat{f}(L)$	L -approximation of the time series	As f
K_a	Superlevel complex of the field at threshold a	Simplicial complex
β_k	k th Betti number	Integer ≥ 0
$\mathcal{D}_k$	Persistence diagram in dimension k	Set of pairs
W_p	Wasserstein distance between diagrams	≥ 0
$\mathcal{T}_{CO_2}$	Combined phase–topology signature	Composite object

7 Physical model of transport and reaction

7.1 Mass balance in air-filled pores

In a one-dimensional vertical approximation, the balance is

$$\frac{\partial(\varepsilon_g C)}{\partial t} = \frac{\partial}{\partial z} \left(\varepsilon_g D_{\text{eff}} \frac{\partial C}{\partial z} \right) - \frac{\partial(q_g C)}{\partial z} + R_{\text{bio}} - R_{\text{weath}} - R_{\text{diss}}. \quad (13)$$

Here ε_g is the air-filled porosity, R_{bio} is biological production, R_{weath} is geochemical consumption through weathering, and R_{diss} is net transfer to the aqueous phase. In this equation, C is expressed per unit gas volume, the reaction terms per unit bulk volume, and q_g per unit bulk cross-sectional area. Measurements in ppm require conversion before use in a dimensional mass balance. The signs must be consistent with the definitions of flux and the z coordinate.

Effective diffusivity depends on moisture, total porosity ε , air-filled porosity, and tortuosity χ :

$$D_{\text{eff}} = D_g F(\varepsilon_g, \varepsilon, \chi), \quad \varepsilon_g \simeq \varepsilon - \theta. \quad (14)$$

The choice of F requires physical justification: alternative tortuosity models can give systematically different fluxes. At LEO, the relationship should be calibrated for the basalt packing and the conclusions tested against alternative parameterizations. Any imported diffusivity formula must use the porosity convention in (13), avoiding a second porosity factor already included in that formula.

7.2 Harmonic diffusion benchmark

For $q_g = 0$, $R = 0$, constant D_{eff} and constant ε_g , a harmonic boundary forcing in a semi-infinite medium gives

$$C(z, t) = \bar{C} + A_0 \exp(-kz) \cos(\omega t - kz + \phi_0), \quad k = \sqrt{\frac{\omega}{2D_{\text{eff}}}}. \quad (15)$$

Here depth is positive into the medium. The solution provides two independent checks:

$$\ln A(z) = \ln A_0 - kz, \quad (16)$$

$$\phi(z) = \phi_0 - kz. \quad (17)$$

For a linear unwrapped phase profile,

$$D_{\text{eff}} = \frac{\omega}{2k_{\phi}^2}, \quad k_{\phi} = -\frac{\partial\phi}{\partial z}. \quad (18)$$

Agreement between the slopes in (16) and (17) supports a diffusion interpretation. Disagreement can indicate heterogeneity, reactions, advection, variable diffusivity, or an inappropriate frequency band.

Numerical example. With $D_{\text{eff}} = 2 \times 10^{-6} \text{ m}^2/\text{s}$ and period $T = 24 \text{ h}$, $\omega = 2\pi/86400 \text{ s}^{-1}$ and $k \approx 4.26 \text{ m}^{-1}$. The ideal model predicts:

Depth	Lag	$A(z)/A_0$
5 cm	0.81 h	0.81
25 cm	4.07 h	0.34
50 cm	8.14 h	0.12

These values illustrate the model and are not prior estimates for LEO. Equation (18) should not be interpreted as a physical diffusivity estimate when phase is nonmonotonic, a deep signal leads a shallow one, or amplitude- and phase-derived estimates of k disagree.

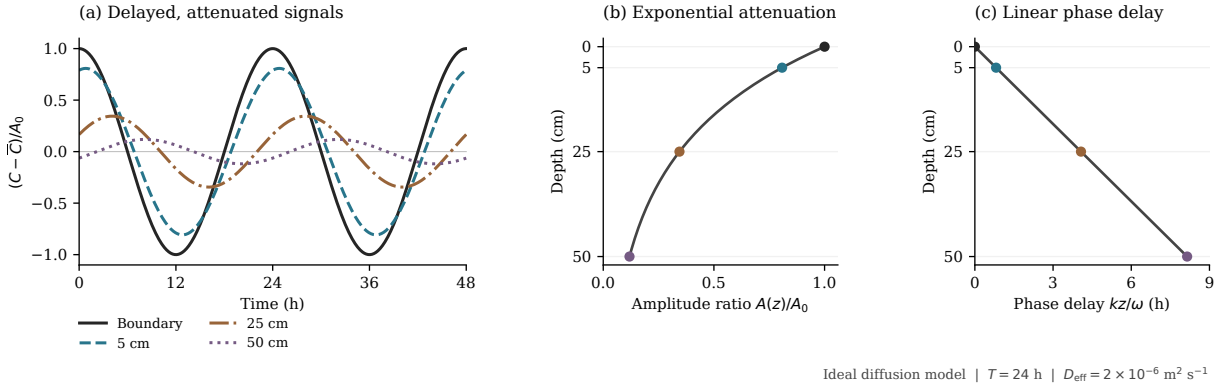


Figure 2: Quantitative harmonic diffusion benchmark, calculated from (15). (a) Normalized concentration anomalies at the boundary and at 5, 25, and 50 cm. (b) Exponential amplitude attenuation with depth. (c) The corresponding frequency-specific phase delay, which increases linearly with depth. All panels use the same daily forcing and diffusivity as the numerical example. The boundary is the ideal surface boundary at $z = 0$, not necessarily the measured air channel. These are model predictions, not LEO observations; joint agreement of measured amplitude and phase profiles is the physical test.

7.3 When the air channel is not the boundary condition

Above-surface concentration is affected by air mixing, the boundary layer, vegetation, pressure, and enclosure ventilation. Two analyses are therefore needed:

$$\text{full column: air} \rightarrow 5 \rightarrow 25 \rightarrow 50 \text{ cm}, \quad (19)$$

$$\text{subsurface column: } 5 \rightarrow 25 \rightarrow 50 \text{ cm}. \quad (20)$$

The subsurface analysis is less exposed to uncertainty in the surface boundary condition. The full-column analysis is needed for atmosphere–substrate exchange but requires a separate surface-resistance model.

8 Data preparation

8.1 Synchronization and quality control

Align all channels to a common time grid t_n . For records sampled at roughly 15-minute intervals, avoid interpolating long gaps. Retain a quality flag $q_{jn} \in \{0, 1, 2\}$ for every observation, denoting invalid, suspect, or usable data. Check:

- physical measurement limits and sensor saturation;
- restart discontinuities, zero drift, and clock offsets;
- responses to condensation and water saturation of the pores;
- flat segments and repeated digital codes;
- availability of concurrent temperature and moisture measurements;
- calibration changes between campaigns.

8.2 Concentration transformation

For positive, heteroscedastic signals, a useful transformation is

$$y_j(t) = \log \frac{C_j(t)}{C_{j,\text{ref}}}, \quad (21)$$

where $C_{j,\text{ref}}$ is the median over a dry baseline period or a physically justified background concentration. Remove a slowly varying trend $b_j(t)$ and scale the anomaly:

$$c_j(t) = \frac{y_j(t) - b_j(t)}{s_j}, \quad (22)$$

where $s_j > 0$ is a robust scale, such as the median absolute deviation. Normalization supports comparisons in multichannel state space; retain absolute concentrations for mass-balance analysis.

8.3 Accounting for measured environmental drivers

To avoid interpreting moisture effects as pedogenesis, fit a conditional model:

$$y_j(t) = f_j[T_j(t), \theta_j(t), \psi_j(t), P_{\text{air}}(t), Q(t), I(t)] + r_j(t), \quad (23)$$

where Q is drainage flux, I is rainfall intensity, f_j is an interpretable regression, generalized additive model (GAM), or physics-informed model, and r_j is the residual. Analyze both y_j and r_j : the former describes the full observed dynamics; the latter describes structure unexplained by the measured drivers. Unexplained structure alone is not evidence of soil formation.

9 Event cycles and temporal scales

9.1 An event coordinate

Divide each rainfall event e into stages:

$$\text{dry} \rightarrow \text{wetting} \rightarrow \text{maximum water storage} \rightarrow \text{drainage} \rightarrow \text{dry-back}. \quad (24)$$

Define a dimensionless event coordinate $s_e \in [0, 1]$ that progresses monotonically through these stages. It allows cycles of different durations to be compared, but must not conceal actual time delays. Perform the analysis in both physical time t and event coordinate s_e .

9.2 Frequency selection

Potentially informative scales include:

- the daily cycle, where it is coherent across several depths;
- the frequency of repeated artificial rainfall;
- wetting-front transit times;
- weekly and monthly scales of baseline evolution.

Inspect spectral power and interchannel coherence before interpreting phase. Exclude phases where the amplitude is near the noise floor or coherence is low.

10 Estimating lags, transfer relationships, and coherence

10.1 Cross-correlation

For channels i, j , define the normalized cross-correlation

$$\rho_{ij}(\tau) = \frac{\mathbb{E}[(c_i(t) - \bar{c}_i)(c_j(t + \tau) - \bar{c}_j)]}{\sigma_i \sigma_j}. \quad (25)$$

Its maximizing lag,

$$\hat{\tau}_{ij} = \arg \max_{\tau \in \Omega_\tau} \rho_{ij}(\tau), \quad (26)$$

provides an initial delay estimate. Choose Ω_τ from physically plausible transport rates. Multiple peaks, trends, periodicity, and local sources can make this estimate ambiguous; it must not be the sole measure of phase relationships.

10.2 Cross-spectral analysis

The Fourier transform resolves a signal into harmonic components:

$$X_j(\omega) = \int_{-\infty}^{\infty} c_j(t) e^{-i\omega t} dt. \quad (27)$$

The modulus $|X_j(\omega)|$ measures the contribution at a frequency, and $\arg X_j(\omega)$ gives its phase. Within an approximately stationary window, estimate the auto-spectra $S_{ii}(\omega)$ and $S_{jj}(\omega)$ and cross-spectrum

$$S_{ij}(\omega) = \mathbb{E}[X_i^*(\omega) X_j(\omega)]. \quad (28)$$

Here the expectation denotes ensemble averaging, estimated in practice by consistent segment, taper, or frequency smoothing; a common spectral-density normalization is understood. An individual product $X_i^* X_j$ is a cross-periodogram. Using an unsmoothed single product with its corresponding auto-periodograms would make magnitude-squared coherence identically one wherever both are nonzero.

The relative phase and magnitude-squared coherence are

$$\Delta\phi_{ij}(\omega) = \arg S_{ij}(\omega), \quad (29)$$

$$\gamma_{ij}^2(\omega) = \frac{|S_{ij}(\omega)|^2}{S_{ii}(\omega)S_{jj}(\omega)}. \quad (30)$$

Under this sign convention, the frequency-specific phase delay is

$$\tau_{ij}(\omega) = -\frac{\text{unwrap } \Delta\phi_{ij}(\omega)}{\omega}. \quad (31)$$

Check the sign against a synthetic signal with a known delay. Unwrap across frequency and depth only over connected regions of high coherence. This quantity is a phase delay, not automatically a unique transit time or a group delay.

Example. If $c_i(t) = \cos \omega t$ and $c_j(t) = \cos[\omega(t-\tau)]$, convention (28) gives $S_{ij} \propto e^{-i\omega\tau}$. Hence $\Delta\phi_{ij} = -\omega\tau$, and (31) recovers a positive delay on the correct branch. Coherence close to one indicates a stable linear relationship at that frequency; it does not establish causality.

10.3 Wavelet phase

A wavelet is an oscillatory function localized in time and scale. The continuous wavelet transform is

$$W_j(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} c_j(t) \psi^*\left(\frac{t-b}{a}\right) dt, \quad (32)$$

where ψ is the mother wavelet, a is a scale related to the characteristic period, b is the time location, and the asterisk denotes complex conjugation. Small $|a|$ resolves faster variations; large $|a|$ resolves slower ones. Unlike a global Fourier spectrum, the transform identifies when a scale is present.

For nonstationary responses, use complex coefficients $W_j(a, t)$. Interpret the local relative phase

$$\Delta\phi_{ij}(a, t) = \arg [W_i^*(a, t)W_j(a, t)] \quad (33)$$

only outside the cone of influence and where wavelet coherence is statistically significant. This is particularly useful for individual rainfall pulses and responses whose duration changes over time.

Example. During the first 12 h after rainfall, the strongest coupling between 5 and 25 cm may occur at scales of 6–10 h; during drying, it may shift to 1–3 days. A global spectrum combines these regimes. A wavelet map locates each regime in time and identifies its local phase relationship.

11 The complex phase state

11.1 Analytic signals

For a real signal $x(t)$, the Hilbert transform is

$$\mathcal{H}[x](t) = \frac{1}{\pi} \text{PV} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau, \quad (34)$$

where PV denotes the Cauchy principal value, treating the singularity at $\tau = t$ by a symmetric limit. In frequency space, positive and negative frequencies receive opposite quarter-cycle phase shifts. Under the convention used here, $\mathcal{H}[\cos(\omega t)] = \sin(\omega t)$ for $\omega > 0$.

For a selected narrowband component $c_j^{(b)}(t)$, define

$$z_j^{(a)}(t) = c_j^{(b)}(t) + i\mathcal{H}[c_j^{(b)}](t) = A_j(t)e^{i\phi_j(t)}. \quad (35)$$

Discard phase estimates where $A_j(t) < A_{j,\min}$: phase becomes unstable near zero amplitude and is undefined at zero.

For transient events, complex wavelet coefficients or complex modes from a dynamic decomposition may be preferable to a global Hilbert transform. Otherwise, a numerically smooth phase can have little physical meaning.

Example. For $x(t) = A \cos(\omega t + \phi_0)$ with $A > 0$ and $\omega > 0$, the analytic signal is $z^{(a)}(t) = Ae^{i(\omega t + \phi_0)}$. Thus $|z^{(a)}| = A$, and its unwrapped phase is $\omega t + \phi_0$. A single nonoscillatory pulse does not admit an equally direct phase interpretation.

11.2 Relative phases and the phase torus

Multiplying all components by a common factor $e^{i\alpha(t)}$ leaves their relative configuration unchanged. A four-level column therefore requires only three relative-phase coordinates:

$$\Phi_m(t) = \begin{bmatrix} \phi_{m1} - \phi_{m0} \\ \phi_{m2} - \phi_{m1} \\ \phi_{m3} - \phi_{m2} \end{bmatrix} \pmod{2\pi}. \quad (36)$$

Each coordinate lies on a circle S^1 , so

$$\Phi_m(t) \in S^1 \times S^1 \times S^1 = \mathbb{T}^3. \quad (37)$$

A rainfall-cycle trajectory may close and wind around one or more torus coordinates. Closed trajectories in different regimes can have different homotopy classes. A continuous deformation through valid closed trajectories cannot change that class; loss of closure or of a well-defined phase must therefore be examined when interpreting apparent changes.

Example. If the unwrapped 5–25 cm phase difference increases by 2π over a closed cycle, while the other differences return without a full turn, the winding vector is $(0, 1, 0)$ in the coordinate order air–5, 5–25, and 25–50 cm. This describes a circuit in phase space and does not imply a physical cavity in the substrate.

12 The phase-transport graph

12.1 Definition

Define a directed weighted graph

$$G_\omega(t) = (V, E, W), \quad (38)$$

whose vertices represent sensors. Include an edge $i \rightarrow j$ when:

- (a) the sensors are geometric or hydrological neighbors;
- (b) their coherence γ_{ij}^2 exceeds a prespecified threshold;
- (c) the lag sign is stable under bootstrap resampling of windows.

One possible weight is

$$w_{ij}(\omega, t) = \gamma_{ij}^2(\omega, t) \exp \left[-\frac{|\tau_{ij}(\omega, t)|}{\tau_0} \right], \quad (39)$$

with direction from the leading to the lagging channel. Choose the scale τ_0 before analysis or derive it from the reference diffusion model.

12.2 Graph descriptors

Compute:

- the number of connected components, specifying weak or strong connectivity;
- vertical and lateral connectivity;
- mean path length from the air or 5 cm layer to 50 cm;
- node centralities that may identify sources or transport junctions;
- stability of edge directions across repeated cycles;
- alternative paths and communities with synchronized phases.

Specify whether path length counts edges or uses a defined edge cost; the coherence-based weight is a similarity, not itself a distance. Build candidate edges primarily from sensor geometry. A graph based solely on correlations can connect remote nodes responding to a shared forcing and invite a false transport interpretation.

Example. The sequence air→5→25→50 cm with increasing delays is consistent with downward propagation. An edge 5→50 cm accompanied by weak 5→25 cm coupling suggests a preferential pathway, subject to ERT, moisture, or tracer validation. A persistent edge 50→25 cm may indicate a deep source or an incorrect phase branch; independent data are needed to distinguish them.

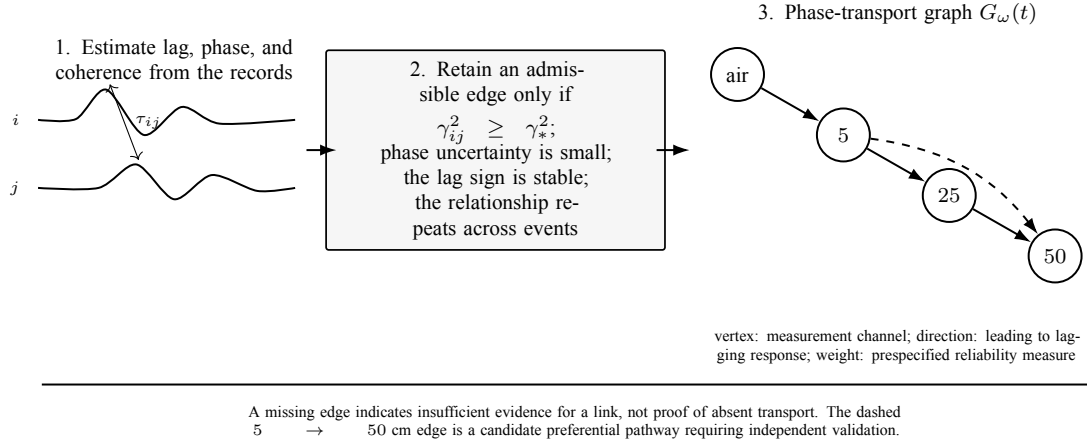


Figure 3: From time series to a phase-transport graph. The middle block applies statistical criteria: visual similarity alone does not establish an edge. Candidate edges must also satisfy the physical-neighbor criterion.

13 Buchstaber’s time-series unfoldings

13.1 Sources and role in the framework

This component follows V. M. Buchstaber’s multidimensional unfolding method, introduced in 1997 and developed further in 2025 [3, 4]. The later work relates unfoldings to Kolmogorov widths and extremal problems on Grassmann manifolds, develops a q -width, Schubert-basis regression, and L -approximation, and includes an atmospheric CO_2 example.

Unfolding is a geometric construction, not a topological phase or a replacement for spectral, wavelet, or Hilbert analysis. It maps a record to a curve in delay-coordinate space, quantifies its effective dimensionality, identifies an informative linear subspace, and reconstructs an approximating time series. Physical lags and phases must then be checked for preservation.

13.2 Step B1: unfold the record

Let

$$f = (f_1, \dots, f_N), \quad f_k = f(t_0 + (k-1)\Delta t) \quad (40)$$

be a regularly sampled record after quality control. In this section, N denotes record length, rather than the number of network channels. A window of length n , $1 < n < N$, yields $P = N - n + 1$ overlapping vectors:

$$X_k = (f_k, \dots, f_{k+n-1})^\top \in \mathbb{R}^n, \quad k = 1, \dots, P. \quad (41)$$

Joining consecutive vectors by line segments produces the oriented piecewise-linear curve

$$X_f = X_f(N; n) \subset \mathbb{R}^n, \quad (42)$$

called the n -dimensional unfolding [3, 4]. Adjacent vectors share $n - 1$ observations, preserving temporal order and local history.

The window spans $(n - 1)\Delta t$. At 15-minute sampling, $n = 97$ spans one day. Choose n from the physical timescales of interest and assess sensitivity to that choice.

LEO example. For $N = 480$ and $n = 97$, the unfolding contains $P = 384$ vectors in $\mathbb{R}^{97}$. A rainfall response occupies a segment of the trajectory of daily windows, rather than a single point.

13.3 Step B2: scatter matrix, width, and effective rank

Define

$$\bar{X} = \frac{1}{P} \sum_{k=1}^P X_k, \quad \tilde{X}_k = X_k - \bar{X}, \quad W_f = \frac{1}{P} \sum_{k=1}^P \tilde{X}_k \tilde{X}_k^\top. \quad (43)$$

The matrix W_f is symmetric and positive semidefinite. Write its eigenvalues as $\lambda_1 \geq \dots \geq \lambda_n \geq 0$, with orthonormal eigenvectors v_j . The subspace

$$L_q^{\text{PCA}} = \text{span}(v_1, \dots, v_q) \quad (44)$$

minimizes the mean squared orthogonal projection error of the centered vectors. Under normalization (43), the normalized q -width for the quadratic componentwise objective used here is

$$d_q(f; n) = \sum_{j=q+1}^n \lambda_j. \quad (45)$$

This is a mean squared residual, not an unsquared worst-case Kolmogorov width. Omitting $1/P$ multiplies the result by P , so comparisons require a consistent normalization.

For positive total variance, the effective ε -rank is the smallest q satisfying

$$\frac{\sum_{j=q+1}^n \lambda_j}{\sum_{j=1}^n \lambda_j} \leq \varepsilon. \quad (46)$$

It measures the dimension needed for a linear approximation of specified accuracy, not the number of physical mechanisms. A constant record requires separate handling because the denominator is zero.

LEO example. If the leading three eigenvalues explain 96% of the variance, the effective rank at $\varepsilon = 0.05$ is at most three. This does not establish three underlying processes: a single nonlinear mechanism may require several linear components.

13.4 Step B3: optimization on the Grassmann manifold

The Grassmann manifold $G(n, q)$ consists of all q -dimensional linear subspaces $L \subset \mathbb{R}^n$. Projection selection can be formulated as an extremal problem on $G(n, q)$, with an explicitly stated information criterion [4]. For mean squared projection error of the centered window vectors, the solution is (44).

A phase-preserving adaptation for LEO could use

$$J(L) = \|f - \hat{f}(L)\|_2^2 + \alpha \sum_{(i,j) \in E} |\text{wrap}(\Delta\phi_{ij} - \Delta\hat{\phi}_{ij}(L))|^2. \quad (47)$$

This objective is proposed here and is not attributed to Buchstaber. For a multichannel implementation, f denotes the stacked records; all terms must refer to the same channels and valid phase samples, with any summation over time or frequency specified explicitly. The coefficient α balances the units and scale of the two terms. Fit it on training events and evaluate on held-out events; $\alpha = 0$ provides a baseline. The reconstruction objective after overlap averaging is distinct from the window-projection objective, so its minimizer need not be the PCA subspace.

LEO example. A small overall reconstruction error may coexist with smoothing of a short post-rainfall delay. Comparing $\alpha = 0$ with $\alpha > 0$ tests whether explicitly preserving phase improves the analysis.

13.5 Step B4: reconstruct an L-approximation

Project the window vectors:

$$Y_k(L) = \bar{X} + \pi_L(X_k - \bar{X}). \quad (48)$$

The overlapping coordinates need not agree after projection. Restore a valid unfolding by projecting back onto the space of consistent window arrays [3, 4]. If $y_{i,k}$ is coordinate i of Y_k , set

$$\hat{f}_s(L) = \frac{1}{|\mathcal{I}_s|} \sum_{(i,k) \in \mathcal{I}_s} y_{i,k}, \quad \mathcal{I}_s = \{(i,k) : i + k - 1 = s\}. \quad (49)$$

Here $1 \leq i \leq n$, $1 \leq k \leq P$, and $1 \leq s \leq N$. Each reconstructed value averages all representations of that sample along an anti-diagonal. The record $\hat{f}(L)$ is the L -approximation, with residual

$$r_f(L) = f - \hat{f}(L). \quad (50)$$

LEO example. For $L = L_3^{\text{PCA}}$, inspect residual autocorrelation and rainfall dependence, and check interdepth lags in the reconstruction. Structured residuals can indicate an unsuitable n or q , or a regime that the linear approximation fails to capture.

13.6 Step B5: extend to multiple channels

A conservative implementation unfolds the air, 5, 25, and 50 cm records separately, using the same Δt , n , and normalization. Compare eigenvalue spectra, effective ranks, q -widths, reconstructions, and residuals.

For joint analysis, define a block vector

$$X_k^{\text{multi}} = [\mathbf{c}^T(t_k), \dots, \mathbf{c}^T(t_{k+n-1})]^T \in \mathbb{R}^{Jn}, \quad (51)$$

where J is the number of channels. This natural multichannel extension should be compared with separate unfoldings because it is sensitive to scaling and missing observations.

LEO example. Four levels with $n = 97$ give 388-dimensional vectors. Assess reliability with block bootstrap resampling and leave-one-level-out analysis.

13.7 Step B6: integrate unfolding into the analysis

Use the method:

1. after synchronization and quality control, on the original, detrended, and residual records;
2. to examine windows spanning rapid rainfall responses, daily variation, and multiday drying;
3. before the final phase analysis, to select q , reconstruct $\hat{f}(L)$, and diagnose residuals;
4. to detect regime changes by comparing d_q , rank, and optimal subspaces in moving windows;
5. to compare matched dry intervals when assessing reversibility;

6. alongside ERT, hydrology, chemistry, mineralogy, and biology for validation.

Compare $L_a, L_b \in G(n, q)$ through their principal angles:

$$d_G(L_a, L_b) = \left(\sum_{\ell=1}^q \vartheta_\ell^2 \right)^{1/2}. \quad (52)$$

This standard Grassmann geodesic distance quantifies a change in the low-dimensional subspace without identifying its cause. Comparisons require the same ambient coordinates, scaling, and dimension q . Near-degenerate eigenvalues at the truncation boundary can make an estimated subspace unstable and should be included in uncertainty analysis.

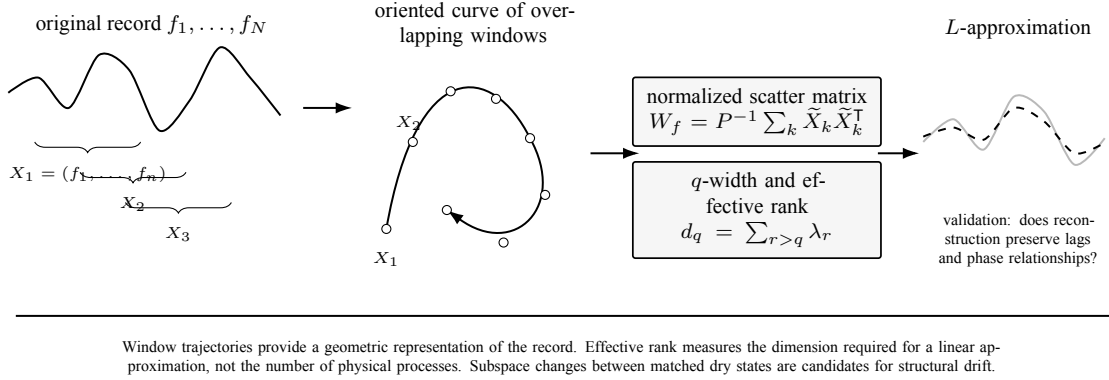


Figure 4: Buchstaber unfolding within the LEO analysis. Overlapping windows define a geometric object, and reconstruction returns it to time-series form for direct lag and phase checks. The normalized scatter matrix and squared-error width follow the definitions in the text.

13.8 Benefits, limitations, and contribution

Unfolding expresses temporal order as curve geometry; the q -width quantifies linear approximation error; a point on $G(n, q)$ is independent of the basis used to represent it; and an L -approximation returns the result to time-series form so that lags and phases can be checked. The atmospheric CO₂ application in the 2025 work provides precedent for the method, although LEO pore-gas dynamics pose a different physical problem [4].

Limitations include dependence between overlapping vectors, sensitivity to n and q , mixing of nonlinear regimes under linear projection, and dominance by slow trends. Address these with block bootstrap resampling, sensitivity analysis, and parallel analyses of original and residual signals.

The proposed contribution is the integration with a depth-resolved diffusion–reaction framework: compare unfoldings across depths and rainfall stages; test q -widths and d_G as descriptors of reversible and residual change; and verify lag and phase preservation after reconstruction. These descriptors enter the combined signature alongside the transport graph, geometric phase, and persistent homology. Until a comprehensive review establishes otherwise, this should be described as a proposed adaptation rather than an absolute priority claim.

14 Geometric phase of an event cycle

14.1 The normalized state

Normalize the network's complex vector:

$$|\psi(t)\rangle = \frac{\mathbf{z}^{(a)}(t)}{\|\mathbf{z}^{(a)}(t)\|_2}, \quad \langle\psi(t)|\psi(t)\rangle = 1. \quad (53)$$

The norm must be nonzero. Normalization removes the common amplitude scale while retaining the distribution of relative amplitudes and phases across sensors. Absolute mass balance remains a separate analysis.

Here $\mathbf{z}^{(a)} = (z_1^{(a)}, \dots, z_N^{(a)})^\top$ contains the analytic signals of the retained channels, and

$$\|\mathbf{z}^{(a)}\|_2 = \sqrt{\sum_{j=1}^N |z_j^{(a)}|^2}. \quad (54)$$

Thus $\langle\psi|\psi\rangle = 1$ specifies a unit norm, not equal sensor concentrations. Use a fixed channel set within a trajectory and across comparisons; changing the vector's coordinates can create artificial geometric changes.

Example. Multiplying every analytic signal by three triples the overall amplitude but leaves $|\psi\rangle$ unchanged. Increasing only the 25 cm signal changes the normalized vector. State geometry therefore responds to redistribution across sensors while ignoring a common positive amplitude scale.

14.2 Discrete holonomy

For samples $t_1, \dots, t_K$ along a closed path, or a path explicitly closed by a shortest geodesic in projective space, define the discrete Pancharatnam–Berry phase

$$\eta_e = -\arg \left[\prod_{k=1}^{K-1} \langle\psi(t_k)|\psi(t_{k+1})\rangle \langle\psi(t_K)|\psi(t_1)\rangle \right]. \quad (55)$$

Every overlap in the product must be nonzero; near-zero overlaps make the estimate unstable. The expression is invariant under an arbitrary common phase factor at each sampled state and describes the path's geometry in projective space. For a sampled smooth path, check convergence as the sampling is refined. An open pulse response must not be described as an observed closed cycle: either document a return to a comparable dry state or state the geodesic closure rule explicitly. Orthogonal endpoints do not define a unique shortest-geodesic closure of this form.

Gauge invariance. Under $|\psi(t_k)\rangle \mapsto e^{i\alpha_k} |\psi(t_k)\rangle$, each neighboring overlap acquires the factor $e^{i(\alpha_{k+1}-\alpha_k)}$. These factors cancel around the closed product. Hence η_e does not depend on the arbitrary common phase assigned at each time.

Example. Two rainfall events may share the same initial and final normalized states. In one, the amplitude maximum passes through 5, 25, and 50 cm in sequence; in the other, the deepest level responds before the middle level. Although their endpoints agree, their overlap products and geometric phases may differ.

14.3 Interpretation

Endpoint concentration differences cannot distinguish all depth-response sequences. The geometric phase η_e and the trajectory on $\mathbb{T}^3$ can retain that information. Slow changes in

$$\Delta\eta_e = \text{wrap}(\eta_e - \eta_{e0}) \quad (56)$$

are candidate indicators of structural evolution only after controlling for moisture, temperature, and the rainfall forcing. Geometric phase is generally a path-dependent geometric quantity, not a quantized topological invariant.

15 Topology of the spatial concentration field

15.1 Superlevel filtration

At each time t , interpolate concentrations onto a physically specified mesh or assign them to the sensor graph. For threshold a , consider

$$K_a(t) = \{\mathbf{r} : C(\mathbf{r}, t) \geq a\}. \quad (57)$$

Use a consistent simplicial or cubical representation of this superlevel set. As a decreases, CO₂-enriched regions appear and merge. Persistent homology records the birth and death thresholds of each feature:

$$\mathcal{D}_k(t) = \{(b_r, d_r)\}_{r}. \quad (58)$$

The persistence $|d_r - b_r|$ helps distinguish robust features from small perturbations, but does not by itself establish a physical origin. State the descending-threshold convention, or equivalently compute a sublevel filtration of $-C$. Sublevel analysis of C can describe depleted regions.

15.2 Interpreting Betti numbers

- β_0 . The number of connected CO₂-enriched domains. An increase may reflect local biological sources or gas pockets; a decrease may reflect merging or gas displacement by water.
- β_1 . The number of independent one-dimensional cycles at a given threshold. Persistent cycles may suggest routes around poorly conducting regions, but require cautious interpretation on a sparse network.
- β_2 . The number of two-dimensional homology classes, interpreted as enclosed voids in a suitable three-dimensional field. A network of 48 sensors will generally require a strong spatial model and extensive sensitivity analysis to support such an inference.

Betti numbers count features at a threshold; persistence measures their lifetime across thresholds. A one-dimensional sensor graph alone cannot represent three-dimensional cavities.

Compare times using, for example,

$$D_{\text{top}}(t_1, t_2) = W_2[\mathcal{D}_k(t_1), \mathcal{D}_k(t_2)]. \quad (59)$$

Specify the diagram metric, diagonal matching, and treatment of essential classes. Repeat the calculation over an ensemble of admissible interpolations so that uncertainty between sensors is not mistaken for physical topology.

Example. At a high threshold, three separate enriched regions give $\beta_0 = 3$. When two merge as the threshold falls, β_0 becomes 2. A short-lived component may be noise; a physically persistent source may generate a feature over a broader threshold range, subject to independent validation.

16 Topology of temporal trajectories

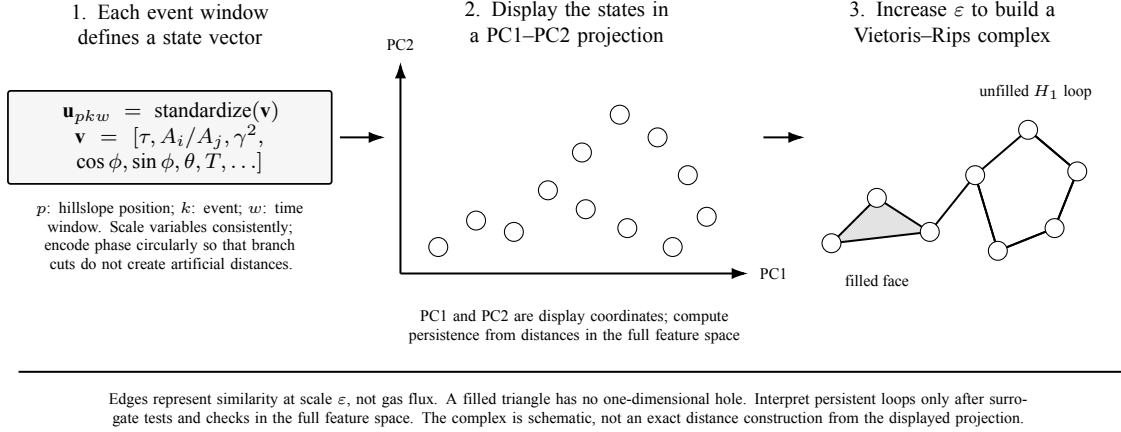


Figure 5: Constructing a topological summary of LEO trajectories. Vertices here represent event-window states; vertices in the phase-transport graph (Figure 3) represent measurement channels.

Alongside spatial topology, analyze the geometry of the temporal trajectory. After a physically interpretable dimension reduction, such as PCA of standardized channels, construct

$$\mathcal{X}_e = \{\mathbf{s}(t_n) : t_n \in \text{event } e\}, \quad \mathbf{s}(t_n) \in \mathbb{R}^d. \quad (60)$$

Build a Vietoris-Rips filtration on this point cloud. Persistent H_1 provides evidence of loop-like geometry. Temporal order and physical covariates are still needed to establish a dynamic cycle or hysteresis, since the point cloud alone discards ordering. Distinguish:

- a closed loop, consistent with a repeatable cycle and potentially with hysteresis;
- an approximately retraced path, consistent with a nearly reversible response;
- an open trajectory with residual drift, indicating failure to return to the starting state;
- distinct trajectory branches under similar external conditions, suggesting multiple regimes.

Dimension reduction can create or remove apparent loops. Check homology in the original state space using a well-defined metric and an explicit treatment of missing data and channel correlations.

Example. If concentration and water storage follow different paths during wetting and drying, the state cloud may form a loop with a persistent H_1 class. If the final dry state remains far from the initial one, describe the result as residual drift rather than a completed cycle.

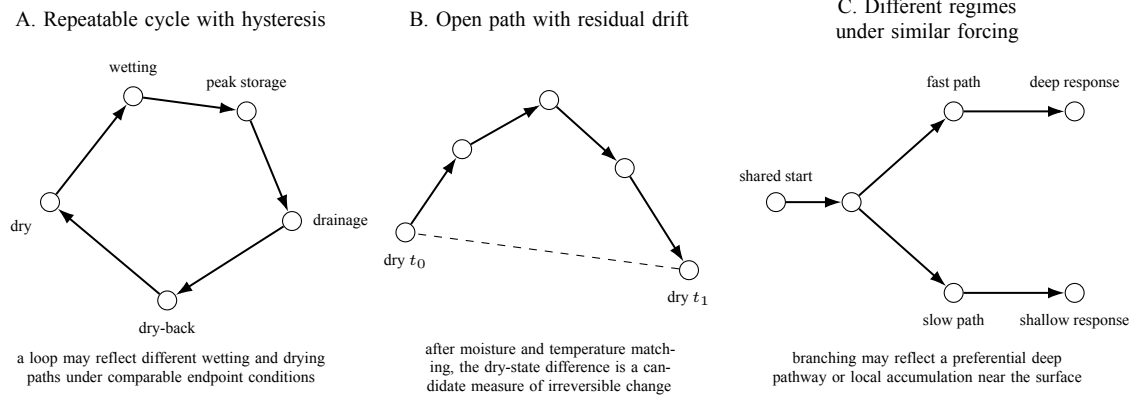


Figure 6: Candidate environmental interpretations of trajectory geometry. These are testable conceptual models, not empirical findings from LEO. Event time determines direction; position in the drawing does not.

17 The combined phase–topology signature

Retain the descriptors as a structured object rather than aggregating them prematurely:

$$\mathcal{T}_{CO_2}(e) = \left\{ \tau_e, \Delta\phi_e, \gamma_e, \mathbf{k}_{\phi,e}, \hat{\mathbf{D}}_{\text{eff},e}, G_e, \mathbf{d}_{q,e}, \mathbf{L}_e, \mathbf{r}_e, \boldsymbol{\eta}_e, \mathcal{D}_e^{\text{space}}, \mathcal{D}_e^{\text{traj}} \right\}. \quad (61)$$

The vector $\mathbf{d}_{q,e}$ contains widths for selected separate and joint unfoldings; $\mathbf{L}_e$ contains optimal subspaces or basis-independent distances between them; and $\mathbf{r}_e$ summarizes reconstruction residuals. These components have different physical meanings. Their identifiability must be assessed before aggregation. If an application requires a scalar classifier, fit it on training data with an independent measure of soil development, for example:

$$I_{\text{top}} = w_1 D_{\text{dry}} + w_2 P_0 + w_3 P_1 + w_4 |\Delta\eta| + w_5 H_{\text{cycle}}. \quad (62)$$

Here D_{dry} compares topology at matched dry states, P_k is total statistically supported persistence in dimension k , and H_{cycle} measures hysteresis. Define and scale each component explicitly. Estimate the weights w_i with cross-validation, not by adjusting them after seeing the desired outcome.

Example. Two events with similar mean CO_2 can differ in τ_e , G_e , and $\mathcal{D}_0^{\text{space}}$. The structured signature preserves these distinctions. Reducing it to one number before independent calibration can obscure mechanisms and encourage overfitting.

18 Distinguishing candidate mechanisms

The following patterns guide interpretation; none uniquely identifies a mechanism without the independent checks listed.

Regime	Phase pattern	Spatial/topological pattern	Independent check
Downward diffusion	Lag increases monotonically with depth; high coherence	Vertically connected domain with progressive attenuation	Agreement between phase- and amplitude-derived D_{eff}
Deep production	Deep signal leads the shallow signal; nonmonotonic phase profile	Persistent local CO_2 maximum	Biomass, organic C, temperature
Geochemical consumption	Amplitude suppression after wetting; possible additional phase shift	CO_2 depletion follows the wetting front	DIC, pH, Ca, Mg, Na, Si; C balance
Preferential pathway	Short lag between widely separated depths	Rapid establishment of connectivity	ERT, moisture, tracer
Isolated gas domain	Low coherence and slow relaxation	Separate persistent component	ERT and local air-filled porosity
Gas displacement by water	Nearly simultaneous sharp pulse inconsistent with diffusion	Simultaneous disappearance of several domains	Increased water storage and pore pressure

19 Separating reversible responses from residual change

19.1 Matched dry states

For each event, select pre- and post-event states with closely matched covariates:

$$|\theta_{\text{pre}} - \theta_{\text{post}}| < \delta_{\theta}, \quad |T_{\text{pre}} - T_{\text{post}}| < \delta_T, \quad |P_{\text{pre}} - P_{\text{post}}| < \delta_P. \quad (63)$$

The residual distance

$$\Delta_{\text{irr}} \mathcal{T}_e = d(\mathcal{T}_{\text{post,dry}}, \mathcal{T}_{\text{pre,dry}}) \quad (64)$$

is a candidate measure of irreversible change, with d defined for the selected signature components. It can still reflect instrument drift or unmeasured covariates. Compare it against control sensors and hillslope replicates before assigning a physical cause.

19.2 Hierarchical decomposition

For descriptor Y_{me} , consider

$$Y_{me} = \alpha_m + \beta_1 \theta_{me} + \beta_2 T_{me} + \beta_3 I_e + g(\text{age}) + u_e + \epsilon_{me}, \quad (65)$$

where α_m is a position effect, u_e an event-level random effect, and g a slowly varying nonlinear trend. Moisture-by-age interactions test whether the response to comparable wetting changes as the substrate develops. Circular descriptors require an appropriate circular model or representation rather than direct linear regression on wrapped angles.

20 Statistical validation and uncertainty

20.1 Sources of uncertainty

Account for:

1. CO₂ sensor error and drift;
2. missing observations and differences in sampling resolution;
3. uncertain coordinates and effective sampling volumes;
4. phase-unwrapping ambiguity;
5. frequency-band or wavelet-scale selection;
6. interpolation between sparse observations;
7. differences in rainfall intensity and event shape;
8. multiple testing across frequencies, depths, and events.

20.2 Validation procedures

- Use temporal block bootstrap resampling to retain autocorrelation.
- Resample events and spatial columns at appropriate dependence scales.
- Use phase-randomized surrogates to test a specified null hypothesis about cyclic structure.
- Permute sensor coordinates to test spatial organization where the null's exchangeability assumptions are justified.
- Include rain-free intervals as negative controls.
- Generate synthetic observations from (13) with known parameters.
- Test sensitivity to coherence thresholds and interpolation choices.
- Separate development, calibration, and external-validation data.

Phase confidence intervals must be circular. Compute a weighted circular mean as

$$\bar{\phi} = \arg \left(\sum_{n=1}^L w_n e^{i\phi_n} \right), \quad (66)$$

with nonnegative weights that may reflect amplitude and coherence. The mean is undefined when the resultant vanishes and unstable when it is small; report concentration of the angles as well as their mean.

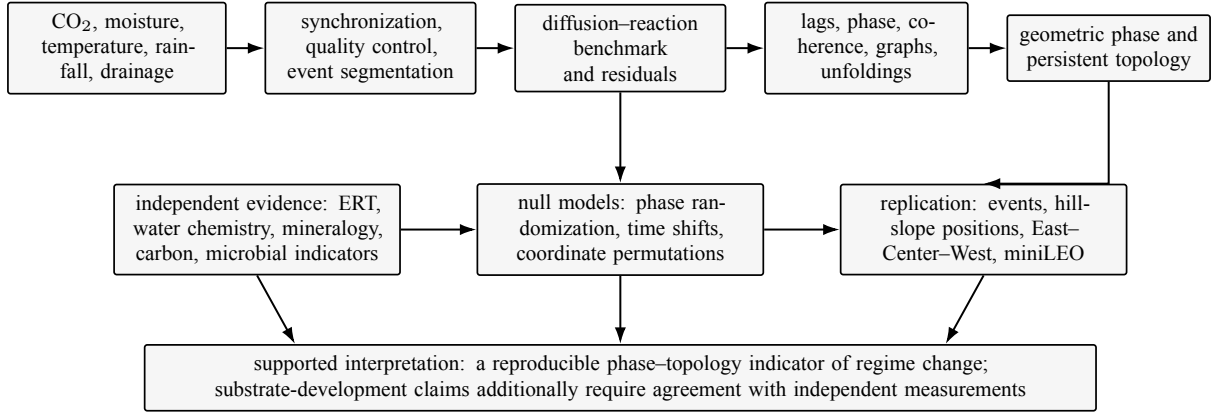


Figure 7: The chain of evidence. Environmental interpretation of topology requires physical checks, null comparisons, independent validation, and replication.

21 Reproducible analysis protocol

Step 1: inventory

Create a channel table recording identifier, hillslope, coordinates, depth, sensor model, units, calibration, operating period, and neighboring sensors. Fix the sign convention for the vertical coordinate.

Step 2: synchronization

Align CO₂, temperature, moisture, water potential, rainfall, gravimetric water storage, drainage, pressure, and ERT on a consistent timeline. Do not fill long gaps for phase analysis.

Step 3: quality control

Apply physical limits, drift diagnostics, flat-signal tests, and condensation flags. Preserve the reason for each exclusion in the quality mask.

Step 4: event segmentation

Identify rainfall onset, wetting-front arrival, maximum water storage, drainage onset, and return to a dry state. Retain physical time and the event coordinate s_e .

Step 5: scale decomposition

Separate the slow trend, daily component, event-scale response, and fast noise. Apply consistent filters to compared channels and verify their phase response.

Step 6: pairwise lags

Estimate $\rho_{ij}(\tau)$, $S_{ij}(\omega)$, $\gamma_{ij}^2(\omega)$, and wavelet phase for air–5, 5–25, and 25–50 cm pairs and lateral neighbors.

Step 7: diffusion benchmark

Fit phase and log amplitude against depth and compare the resulting D_{eff} estimates. Flag non-monotonic phase, leading deep signals, and unusually rapid propagation.

Step 8: phase graph

Construct $G_{\omega}(t)$ from physically admissible neighbors and significant coherent relationships. Assess connectivity, directionality, centrality, and community stability.

Step 9: Buchstaber unfoldings

Build separate and joint unfoldings for several physically motivated window lengths n . Estimate q -widths, effective ranks, and L -approximations. Test lag and phase preservation, block-bootstrap stability, and sensitivity to sensor omission. Compare optimal subspaces of matched dry states on $G(n, q)$.

Step 10: geometric phase

Construct $|\psi(t)\rangle$ from coherent narrowband components and estimate η_e for completed cycles. Check overlap magnitudes, sensitivity to individual sensor omissions, and any alternative path-closure rules.

Step 11: spatial persistence

Compute $\mathcal{D}_0$ and, where sampling density supports it, $\mathcal{D}_1$ for the concentration and residual fields. Repeat across an ensemble of interpolations.

Step 12: trajectory topology

Test for repeatable cycles and hysteresis in state space. Relate trajectory shape and temporal ordering to rainfall stages.

Step 13: reversibility

Compare matched dry states using (64). Separate the rapid reversible response from slow residual drift.

Step 14: physical validation

Compare descriptors with ERT, pore-water and drainage chemistry, DIC, pH, Ca, Mg, Na, Si, mineralogy, organic carbon, and microbial measurements.

Step 15: replication

Assess transfer across East, Center, West, and miniLEO. Spatial sensors must not be treated as fully independent replicates.

22 Symbolic and numerical checks in Wolfram Mathematica

The accompanying script, `verify_phase_topology_CO2_LEO.wl`, separates algebraic consistency from physical applicability. The script was rerun with the local Wolfram kernel for this English edition and completed all eleven implemented checks without a failure. The checks cover:

1. substitution of (15) into the one-dimensional diffusion equation, yielding a zero symbolic residual;
2. recovery of the original D_{eff} from the phase slope using (18);
3. numerical lags and attenuation at 5, 25, and 50 cm for a daily cycle;
4. verification of $\Delta\phi = -\omega\tau$ for the cross-product convention $X_i^* X_j$;
5. verification of $\cos\theta + i\sin\theta = e^{i\theta}$ and the harmonic analytic signal;
6. invariance of the closed product in (55) under $|\psi_k\rangle \mapsto e^{i\alpha_k}|\psi_k\rangle$;
7. geometric phase on a synthetic trajectory and winding number for a full circuit;
8. unfolding node count and exact agreement of overlapping samples;
9. equality of the mean squared optimal projection residual and the sum of discarded scatter-matrix eigenvalues;
10. exact reconstruction by anti-diagonal averaging of a consistent unfolding.

The rerun gave a zero diffusion-equation residual and exact symbolic recovery of D_{eff} . The gauge-phase error was approximately 8.2×10^{-15} rad; the wrapped geometric-phase error was numerically zero; and the winding number was one. The q -width identity error was 2.22×10^{-16} , with zero overlap and exact-reconstruction errors. These are results for the implemented synthetic examples, not universal error bounds. The numerical daily-cycle lags were 0.81434, 4.07169, and 8.14338 h, consistent with the rounded table above.

Scope of these checks. Symbolic equality verifies internal mathematical consistency. It does not establish that homogeneous diffusion adequately describes LEO. That requires model-residual analysis, independent hydrological and geochemical measurements, repeatability across events, and validation on another hillslope.

23 Recommended publication figures

A minimum figure set would include:

1. hillslope layout and vertical sensor columns;
2. rainfall, moisture, and depth-resolved CO_2 over a representative event;
3. wavelet coherence and relative phase for the 5–25 and 25–50 cm pairs;
4. amplitude and unwrapped phase against depth, with diffusion predictions;
5. phase-transport graphs before, during, and after rainfall;

6. a two-dimensional projection of an event trajectory on the phase torus;
7. persistence diagrams or persistence landscapes;
8. long-term drift in D_{dry} and η_e alongside independent geochemical markers;
9. a matrix of differences across hillslopes and experimental regimes.

24 Evidence required for a soil-development claim

A phase–topology indicator of soil development is supported only if all of the following hold:

1. descriptors are reproducible across comparable rainfall cycles;
2. they are robust to noise, missing data, filtering, and interpolation;
3. their changes cannot be explained solely by moisture, temperature, and boundary CO_2 ;
4. residual drift is present between matched dry states;
5. that drift agrees with at least two independent classes of observations, such as ERT and solution chemistry;
6. results transfer to a different hillslope or time interval;
7. synthetic tests establish identifiability of the claimed mechanism;
8. where unfolding is used, conclusions are robust to n and q , and the accepted L -approximation preserves lags and phases within a prespecified tolerance.

Without this evidence, the defensible claim is a change in the topology of the distributed CO_2 field, not direct observation of basalt becoming soil.

25 A two-year PhD research program

25.1 Purpose and assumptions

The following 24-month plan turns the framework into a staged research program. Its objective is a defensible chain of inference: documented observations and validated transport physics, followed by reproducible phase, geometric, and topological descriptors, independent validation, and publication. Months are counted from the formal project start.

The program assumes access to LEO records, sensor metadata, and at least some concurrent moisture, temperature, rainfall, drainage, ERT, and solution-chemistry data. Verify access and coverage during the first two months. If a data class is unavailable, remove the causal interpretation that depends on it and narrow the question to the dynamic organization of the CO_2 field. This fallback preserves the mathematical research but limits claims about soil-forming mechanisms.

Six linked objectives organize the work:

- O1.** establish a quality-controlled, documented dataset of depth-resolved CO_2 and covariates;
- O2.** determine which lags and amplitude changes are consistent with diffusion–reaction transport and which require additional mechanisms;

- O3.** develop a robust phase description of vertical and lateral sensor relationships;
- O4.** use Buchstaber unfoldings to assess effective dimensionality, compare regimes, and reconstruct physically testable records;
- O5.** determine whether geometric phase and persistent topology add information beyond concentrations, lags, and moisture;
- O6.** validate transfer across events, periods, and hillslopes, and prepare the thesis and papers.

Their dependencies are

$$O1 \longrightarrow O2 \longrightarrow \begin{cases} O3, \\ O4 \end{cases} \longrightarrow O5 \longrightarrow O6. \quad (67)$$

Topology needs the physical foundation provided by O1 and O2. Once a common event dataset is available, unfolding and phase analysis can proceed in parallel, using identical training and validation intervals.

25.2 Work packages

Each package produces a result and supplies data or criteria to the next.

Package	Objective	Activities	Verifiable output
WP0	Project management	Data plan, decision log, version control, supervision, quarterly risk review	Traceable decisions and current plan
WP1	Data and metadata	Inventory channels, units, coordinates, calibration, gaps, and configuration changes	Frozen dataset and data dictionary
WP2	Quality control and events	Quality masks, synchronization, drift checks, rainfall cycles, matched dry states	Validated event catalog
WP3	Transport physics	Correlation lags, cross-spectra, wavelet coherence, D_{eff} , residuals	Transport-regime classification with uncertainty
WP4	Phase and graphs	Phase gradients, directed links, graph stability, geometric phase of completed cycles	Phase-transport descriptors and null tests
WP5	Buchstaber unfoldings	Select n, q ; scatter matrices, widths, $G(n, q)$, L -approximations	Geometric descriptors and lag-preservation checks
WP6	Persistent topology	Spatial filtrations, trajectory topology, interpolation sensitivity	Robust persistence diagrams or a negative result
WP7	Integration and validation	Reversibility, mixed models, ERT, water, chemistry, mineralogy, biology, hillslope transfer	Causal interpretation bounded by the evidence

Package	Objective	Activities	Verifiable output
WP8	Papers and thesis	Figures, manuscripts, code archive, thesis chapters, presentations, defense	Reproducible research package

A negative result in WP6 is scientifically valid. If topological descriptors do not outperform simple baselines in robustness or validation, exclude them from the final indicator. The thesis must remain viable without a predetermined positive result.

25.3 Months 1–3: scope, data access, and reproducibility

Objective. Establish the available observations and turn the overall idea into testable questions, variables, and analysis rules. The principal risk is beginning advanced analysis before resolving data provenance and quality.

Activities.

1. Agree with supervisors on the primary and secondary questions, permissible causal claims, and minimum viable thesis contribution.
2. Obtain the data and document usage rights, citation requirements, redistribution restrictions, and archiving obligations.
3. Register each sensor’s hillslope, position, depth, serial number, model, units, sampling interval, operating periods, calibrations, and known interventions.
4. Map the correspondence between CO₂, rainfall, moisture, temperature, water potential, drainage, ERT, chemistry, and biological observations.
5. Define three levels of inference: field description, transport-mechanism interpretation, and substrate transformation. List the independent evidence required at each level.
6. Set up immutable raw data, processed tables, analysis scripts, tests, outputs, and file-level provenance tracking.
7. Undertake focused training in porous-media gas transport, spectral and wavelet analysis, circular statistics, Grassmann manifolds, time-series unfoldings, and persistent homology.

Deliverables. A concise research protocol, data dictionary, sensor map, covariate-availability matrix, risk register, and a test dataset spanning several weeks that can be read and visualized through the complete workflow.

Gate G1. Proceed when the time zone, units, depth convention, channel correspondence, and at least one common operating interval have been established. Otherwise, use months 3–4 to recover metadata with the LEO team. Exclude unverified channels rather than infer their metadata.

25.4 Months 4–6: quality control and the event catalog

Objective. Produce an analysis dataset with traceable exclusions and consistent definitions of rainfall events and dry states.

Activities.

1. Align records without interpolating long gaps; retain original timestamps and a transformation mask.
2. Flag physical outliers, communication failures, flat signals, baseline shifts, possible condensation, and post-maintenance periods.
3. Compare neighboring sensors and depths without automatically treating disagreement as error: it may reflect a local source or heterogeneity.
4. Define rainfall onset and end, wetting-front arrival, peak water storage, drainage, and return to a dry state.
5. Catalog event intensity, duration, antecedent moisture, temperature, channel coverage, and quality class.
6. Split events into development, calibration, and validation sets before the main analysis, documenting the rules before selecting final thresholds.
7. Generate synthetic records with known delays, attenuation, gaps, and drift to test the full computational chain.

Deliverables. Dataset version 1.0, a quality report, an atlas of representative and problematic events, a fixed data split, and automated checks for swapped channels, unit changes, and non-monotonic timestamps.

Gate G2. The dataset must contain enough events with simultaneous observations at at least three levels to obtain stable block-bootstrap uncertainty intervals. If event coverage is insufficient, analyze continuous daily and seasonal variation and label event-based findings exploratory.

25.5 Months 7–9: lag physics and the diffusion benchmark

Objective. Establish a simple physical baseline against which the added value of phase, topology, and unfolding can be tested.

Activities.

1. Estimate correlation lags and confidence intervals for air–5, 5–25, and 25–50 cm pairs and lateral neighbors.
2. Compute cross-spectra, coherence, and phase; verify lag signs on synthetic data and test detrending sensitivity.
3. Apply wavelet analysis to nonstationary rainfall responses, excluding the cone of influence and low-coherence regions.
4. Fit depth-dependent amplitude attenuation and phase gradients, independently estimating and comparing D_{eff} .
5. Test residual dependence on moisture, temperature, rainfall intensity, pressure, and event stage.

6. Classify responses as consistent with downward diffusion, potentially affected by surface resistance, suggestive of a local source, potentially advective or displacement-driven, or unidentifiable.

Deliverables. Lag and phase estimates with uncertainty, maps of departures from the diffusion benchmark, validated frequency bands, and a draft thesis chapter on transport physics.

Gate G3. Reject phase interpretation in bands where apparent repeatability occurs only at low coherence or the sign changes under small filter adjustments. Base WP3 on event lags and transport modeling where a defensible phase cannot be assigned.

25.6 Months 10–12: network analysis and the first paper

Objective. Move from pairwise relationships to network organization and complete a self-contained, verifiable result in year one.

Activities.

1. Restrict graph edges to physically admissible neighbors satisfying prespecified coherence, lag-uncertainty, and repeatability criteria.
2. Compare dry, wetting, drainage, and drying graphs through connectivity, directionality, centrality, and community stability.
3. Apply phase randomization, spatial-coordinate permutations, and rain-free controls under their stated null assumptions.
4. Test on validation data whether graph descriptors add information beyond moisture and mean CO₂.
5. Prepare P1 on depth-resolved lags, the diffusion benchmark, and the phase-transport network. Include topological claims only if their validation is complete.
6. Present at an internal seminar and conduct a formal annual review of questions, negative results, code quality, publications, and the feasibility of year two.

Deliverables. A frozen year-one method version, reproducible figures, a complete first manuscript or preprint, and a written committee decision defining year-two scope.

Gate G4. By month 12, one complete line of evidence must stand independently of persistent topology. If the graph is unstable, restrict P1 to lag physics and regime diagnostics; this need not delay the unfolding work.

25.7 Months 13–15: Buchstaber unfoldings

Objective. Test whether stable low-dimensional subspaces describe CO₂ dynamics and whether their geometry changes across event stages and substrate ages.

Activities.

1. Select a range of window dimensions n for each physically motivated timescale.
2. Unfold the air, 5, 25, and 50 cm records separately, then construct a standardized multichannel unfolding.
3. Estimate scatter matrices, eigenvalue spectra, q -widths, and effective ranks, with block-bootstrap uncertainty.
4. Compare optimal subspaces $L \in G(n, q)$ across event stages and matched dry states through principal angles.
5. Reconstruct L -approximations and directly test preservation of amplitudes, interdepth lags, phase signs, and event-stage ordering.
6. Compare the PCA baseline with the proposed phase-preserving objective. Select α using development data and assess improvement on held-out events.
7. Compare unfolding descriptors against autocorrelation, spectral power, PCA of nonoverlapping features, and a moisture-only model.

Deliverables. An atlas of unfoldings and spectra, justified ranges for n and q , distances between regimes, reconstructed records, and a dedicated report on physical-phase preservation.

Gate G5. The method is useful if conclusions persist over reasonable n, q ranges, block resampling, and omission of one depth, while reconstructed lags remain within the original uncertainty bounds. Otherwise, retain unfolding as a diagnostic appendix rather than a primary transformation indicator.

25.8 Months 16–18: geometric phase and persistent topology

Objective. Test for robust whole-cycle and spatial-field descriptors that add information beyond individual lags or mean concentrations.

Activities.

1. Construct normalized complex states only from channels and bands with validated coherence.
2. Estimate geometric phases for completed cycles and test sensitivity to closure, sensor omission, and amplitude normalization.
3. Compute spatial superlevel filtrations and H_0 ; pursue H_1 only where observation geometry supports it.
4. Repeat persistence analysis across admissible interpolations and for residual fields adjusted for moisture and temperature.
5. Use temporal embeddings for trajectory analysis and compare observed diagrams with phase-randomized and other surrogate records.
6. Test incremental validation performance against the WP3 physical baseline and WP5 geometric descriptors.

Deliverables. Geometric phases and persistence diagrams with uncertainty, plus a written assessment of which descriptors pass null and sensitivity tests. Complete and submit P2 during this quarter alongside the initial P3 analyses.

Gate G6. Include a descriptor in the combined signature only if it is reproducible across comparable events and robust to interpolation, noise, and threshold choice. Failure to detect robust H_1 does not establish an absence of physical structure; sparse sampling may be responsible.

25.9 Months 19–21: integration, residual change, and independent validation

Objective. Determine whether the descriptors track slow substrate change or primarily reversible hydrological responses.

Activities.

1. Match pre- and post-event dry states on moisture and temperature and estimate residual differences using (64).
2. Fit hierarchical models separating position, event, moisture, temperature, rainfall intensity, and experiment-age effects.
3. Combine descriptors by prespecified roles: transport, phase organization, unfolding geometry, spatial connectivity, and residual change.
4. For predictive claims, check temporal precedence: an indicator should not systematically follow the independent outcome it is intended to predict.
5. Compare changes with at least two independent data classes where available, such as ERT and solution chemistry, or mineralogy and biological indicators.
6. Validate on a held-out period or another hillslope without retuning thresholds or signature weights.
7. Examine instrument drift, ventilation changes, temperature, rainfall differences, sensor re-configuration, and unrecorded interventions as alternative explanations.

Deliverables. A final model with uncertainty, a table of supported and rejected hypotheses, a hillslope-transfer assessment, and an explicit statement of the strongest defensible conclusion.

Gate G7. Use “indicator of basalt-to-soil transformation” only if O1–O6 and the requirements in Section 24 are met. With absent or conflicting independent validation, report reorganization of transport and pore-gas CO_2 dynamics.

25.10 Months 22–24: synthesis, publication, and thesis

Objective. Complete a project in which every figure, result, and claim can be traced to a fixed data version and reproduced.

Activities.

1. Freeze the final dataset and computational environment; rerun the analysis from a clean state.
2. Complete and submit P3 on phase, topology, residual drift, transferability, and independent validation. Address editorial decisions on P1 and P2 while preserving each paper's distinct question and principal figures.
3. Organize the thesis around the evidence: data and physics; lags and phase networks; unfoldings; geometric phase and topology; reversibility and validation; limitations.
4. Have another group member reproduce the analysis from written instructions.
5. Prepare appendices containing definitions, derivations, sensitivity analyses, negative results, and all parameters.
6. Archive permitted data, code, checksums, software versions, result provenance, and scripts for every final figure.
7. Prepare the defense and responses on identifiability, multiple testing, n, q selection, interpolation, the physical meaning of phase, and causal limits.

Deliverables. A thesis, three related but scientifically distinct manuscripts, a reproducible archive, and defense materials.

Gate G8. Completion requires automated reproduction of principal tables and figures, uncertainty and appropriate null tests for the claimed comparisons, and conclusions consistent with the available independent evidence. Computational complexity is not a completion criterion.

25.11 Quarterly schedule and milestones

Quarter	Months	Main work	Milestone
Q1	1–3	Scope, access, metadata, training, infrastructure	M1: approved protocol and validated test dataset
Q2	4–6	Synchronization, QA/QC, events, synthetic tests	M2: dataset 1.0 and frozen split
Q3	7–9	Lags, spectra, wavelets, diffusion benchmark	M3: validated transport estimates
Q4	10–12	Phase graph, null models, P1, annual review	M4: complete first manuscript
Q5	13–15	Unfoldings, widths, $G(n, q)$, reconstruction	M5: robust geometry analysis or documented limitation
Q6	16–18	Complete unfolding study and submit P2; begin geometric phase and persistence	M6: P2 submitted and P3 protocol fixed
Q7	19–21	Residual change, mixed models, external validation	M7: final level of scientific inference

Quarter	Months	Main work	Milestone
Q8	22–24	P3, P1–P2 responses, archive, defense	M8: P3 submitted and reproducible thesis package

25.12 Working rhythm, supervision, and training

A two-year schedule requires continuous writing and regular review:

- weekly: review the analysis log, produce one reproducible result, and record the next decision;
- fortnightly: meet the primary supervisor about data, methods, and blockers;
- monthly: consult jointly with biogeochemistry, hydrology, and mathematical-methods specialists;
- quarterly: reproduce key results from a clean state and review risks;
- from month 4: write methods and results as the analysis develops;
- at least twice: present to an audience outside the immediate research group.

An indicative allocation is 60–65% for analysis and development, 15–20% for reading and mathematical training, 15–20% for writing and visualization, and about 5% for data management. Choose allocations within these ranges that sum to 100%; writing will increase in the final quarter. Include required coursework and teaching before approving milestones. If time becomes constrained, narrow the most ambitious interpretation in P3 while retaining its minimum question about incremental performance and transferability as the third required manuscript.

Training should coincide with application. Prioritize pore-gas transport, environmental time series, and reproducibility in the first half-year; spectral methods, wavelets, circular statistics, and graphs in the second; Grassmann manifolds, Buchstaber unfoldings, and persistent homology in the third; and hierarchical models, evidence integration, and scientific writing in the fourth.

25.13 Publication program

The program requires exactly three papers, separated by scientific question and type of evidence.

1. *P1: physically informed lags and transport regimes.* Data, quality control, event segmentation, interdepth lags, coherence, the diffusion–reaction benchmark, and the phase-transport graph support a physically testable classification of CO₂ transport regimes.
2. *P2: the geometry of Buchstaber unfoldings.* q -widths, subspace comparisons, and L -approximations are evaluated through strict amplitude, lag, and phase-preservation tests. The independent contribution is an assessment of the capabilities and limits of low-dimensional representations of LEO dynamics.
3. *P3: phase, topology, residual drift, and transferability.* Geometric phase, spatial and trajectory persistence, surrogates, matched dry states, and independent validation test whether these descriptors reveal repeatable cycles, regime changes, and residual evolution beyond simple baselines.

Match journals to the demonstrated contribution: hydrology, biogeochemistry, or critical-zone science for P1; time-series analysis, environmental informatics, or applied nonlinear dynamics for P2; and interdisciplinary Earth or environmental science for P3 where independent validation supports it. Journal prestige must not drive stronger causal claims than the evidence allows.

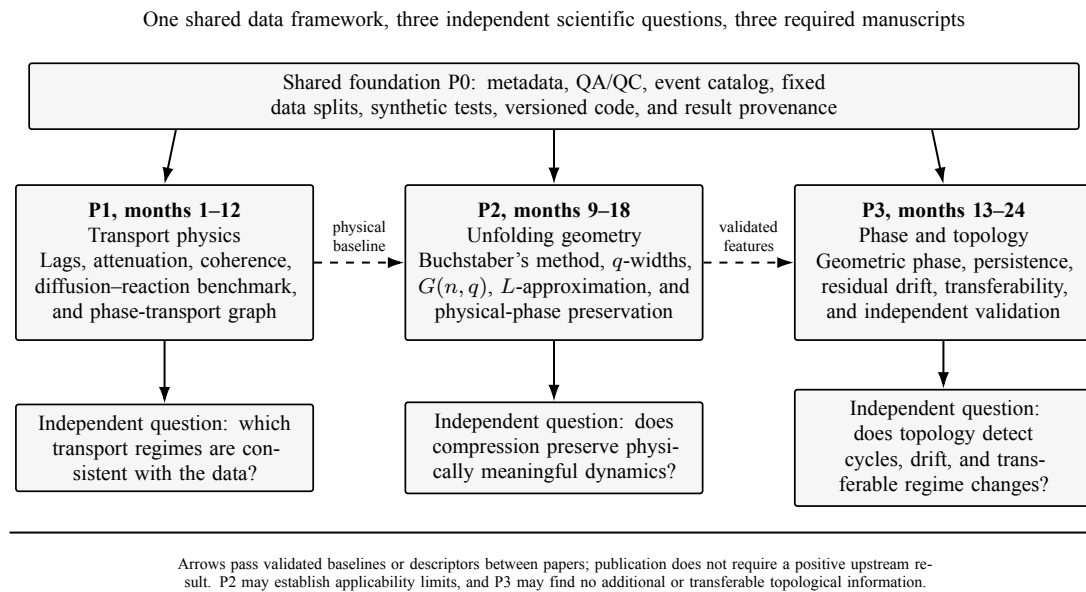


Figure 8: The publication program. A shared reproducible foundation avoids duplicate processing. Distinct questions, principal figures, and readiness criteria keep each paper scientifically independent.

25.13.1 Overall strategy

P1 establishes transport physics. P2 tests whether unfolding compresses multichannel dynamics while retaining physically meaningful lags and phases. P3 asks whether independently validated descriptors capture robust cycles, residual drift, and transfer across events, periods, or hillslopes. Keep P3's central topological claim out of P2, and do not retune unfolding parameters on P3's external-validation data. A negative result in one paper does not invalidate the others' questions.

Before final analysis, specify for each manuscript:

1. one primary question with a possible positive, negative, or conditional answer;
2. the minimum data required for an identifiable answer;
3. the baseline against which the method is evaluated;
4. criteria for including results in the principal figures;
5. the strongest conclusion permitted by the available validation.

Agree on authorship roles in month 1, assigning responsibility for conception, data, software, statistics, physical interpretation, visualization, and editing. Reassess contributions at G4 and G7. Author order should reflect actual work and remain open to revision as contributions develop.

25.13.2 Planned outputs

Code	Focus	Primary question	Minimum result	Submission
P1	Physics and time series	How do lag, attenuation, and propagation direction vary across air and depth?	Validated lags, coherence, diffusion benchmark, uncertainty	Month 12
P2	Time-series unfoldings	Does low-dimensional reconstruction preserve regimes, lags, and physical phase?	Validated reconstruction, n, q robustness, external checks, limitations	Month 18
P3	Phase, topology, and environmental interpretation	Do geometric phase and persistence add reproducible, transferable information on cycles and drift?	Null models, matched dry states, independent validation, transfer assessment	Month 24
P0	Reproducible methods	Can the computational chain be reproduced independently?	Versioned protocol, code, synthetic tests, permitted data	With P1–P3

P0 is supporting research infrastructure, not a fourth planned journal article. Depending on data rights, release it as a repository, supplement, or protocol. It is required for the reproducibility of all three papers.

25.13.3 P1: physically informed analysis of depth-resolved lags

Research question. Can joint lag, phase-gradient, and amplitude-attenuation patterns distinguish downward diffusion from responses requiring surface resistance, local sources, advection, or water-driven gas displacement?

Working title.

Depth-resolved CO₂ lags reveal changing transport regimes in a controlled basalt hillslope.

Revise the title to match the findings. Replace “reveal” or “identify” with “characterize” or “are consistent with” if mechanisms cannot be separated unambiguously.

Manuscript structure.

1. physical problem and LEO measurement geometry;
2. data, synchronization, QA/QC, and event segmentation;
3. correlation lags, cross-spectra, coherence, and wavelet phase;
4. diffusion–reaction benchmark and independent D_{eff} estimates;
5. results across depths, rainfall stages, and positions;
6. alternative mechanisms, uncertainty, and identifiability limits;
7. the phase-transport graph as a network extension of pairwise analysis.

Principal figures.

1. hillslope layout and air, 5, 25, and 50 cm levels;
2. a fully documented rainfall cycle;
3. lag distributions and bootstrap intervals;
4. coherence and phase in prespecified bands;
5. joint amplitude–phase comparison with diffusion;
6. event-stage changes in the phase-transport graph.

Place additional quality checks, events, filter sensitivity, and null analyses in the supplement, while keeping the evidence essential to the main claims in the paper.

Schedule.

- Month 4: create the manuscript outline and a claim–evidence–figure table.
- Month 6: complete data, QA/QC, and event methods.
- Month 8: produce the first full diagnostic figure set.
- Month 9: freeze the primary events and frequency-analysis rules.
- Month 10: write results and discussion while supplements are completed.
- Month 11: obtain coauthor review and independent computational checks.
- Month 12: edit the language, check journal requirements, and submit.

Readiness criterion. The main finding must reproduce on held-out events; lag signs must pass synthetic tests; uncertainty must account for autocorrelation; and alternative explanations must be explicit. A collection of visually striking events does not satisfy these requirements.

25.13.4 P2: unfoldings and low-dimensional geometry

Research question. Do effective rank, q -width, and optimal-subspace position on $G(n, q)$ distinguish regimes robustly, and does L -approximation preserve amplitude, lag, and physical phase after accounting for moisture, temperature, rainfall strength, and conventional time-series descriptors?

Working title.

Multidimensional time-series unfoldings preserve transport structure in distributed CO₂ dynamics of a basalt hillslope.

If reconstruction fails to preserve lags or regime differences, report the resulting applicability limit and revise the title accordingly. Retain the negative result.

Manuscript structure.

1. the limits of individual lags as a description of system organization;
2. Buchstaber's published method and the distinct LEO multichannel adaptation;
3. selection of n, q , widths, principal angles, and L -approximation;
4. the phase-preserving objective and physical-lag preservation tests;
5. geometry across event stages, dry states, and positions;
6. baselines, null hypotheses, and held-out event evaluation;
7. window, dimension, linear-subspace, and mechanism-identifiability limits.

Principal figures.

1. the construction from records to unfolding vectors and reconstruction;
2. eigenvalue spectra and q -widths by regime;
3. principal-angle distances across dry states and rainfall stages;
4. amplitude, lag, and phase preservation after reconstruction;
5. geometric descriptors compared with P1 baselines;
6. regime-classification validation on held-out events.

Schedule.

- Month 10: outline the manuscript and fix baselines.
- Month 12: approve n, q selection rules without consulting validation data.
- Month 14: complete computational checks of unfolding and reconstruction.
- Month 15: freeze robustness analysis and lag/phase preservation criteria.
- Month 16: complete principal figures and write results.
- Month 17: complete the draft, independent reproduction, and methods review.
- Month 18: submit.

Readiness criterion. Clearly separate established theory, the proposed multichannel and phase-preserving adaptation, and the empirical LEO findings. Do not feed validation-set choices back into training. Assess incremental value against prespecified simple baselines.

25.13.5 P3: phase, topology, residual drift, and transferability

Research question. After accounting for moisture, temperature, rainfall strength, and transport physics, do geometric phase and persistent topology identify repeatable cycles, regime changes, and residual drift that generalize to held-out events, periods, or hillslopes?

Working title.

Phase topology distinguishes reversible cycles from residual change in a controlled basalt landscape.

Retain “residual change” rather than “soil formation” until independent ERT, chemical, mineralogical, or biological evidence supports that interpretation. If hillslope transfer is limited, retain the required paper with a narrower question about robustness and transfer limits.

Manuscript structure.

1. distinguishing reversible rainfall responses from residual state change;
2. prespecified events, matched dry states, and covariates;
3. normalized complex states, geometric phase, and admissible-cycle criteria;
4. Vietoris–Rips trajectory filtrations and spatial superlevel filtrations;
5. interpretation of vertices, edges, H_0 components, H_1 loops, and persistence;
6. surrogates and sensitivity to metric, scale, interpolation, and sensor sparsity;
7. incremental validation performance beyond P1 and P2;
8. transfer across events, periods, or hillslopes and independent observations;
9. limits of inference: CO_2 reorganization, transport change, or supported substrate development.

Principal figures.

1. construction of a state trajectory from air–5–25–50 cm records;
2. geometric phase of a completed event and closure uncertainty;
3. state cloud, filtration, and an annotated barcode or persistence diagram;
4. observed topology against surrogate distributions;
5. matched pre- and post-event dry states and residual drift;
6. external validation and comparison with independent data classes.

Schedule.

- Month 13: outline the paper and prespecify its minimum question, metrics, and nulls.
- Month 15: freeze events, dry-state matching, and transfer rules.
- Month 18: complete initial geometric-phase and persistence calculations without outcome-driven threshold changes.
- Month 20: finish sensitivity, surrogate, and incremental-performance analyses.
- Month 21: freeze external validation and the permissible environmental interpretation.
- Month 22: complete the text and principal figures.
- Month 23: conduct scientific, statistical, and reproducibility reviews.
- Month 24: submit.

Readiness criterion. A positive claim requires robustness over justified metrics and parameters, distinction from surrogates, improvement beyond baselines, and validation outside the selection data. Hillslope replication is desirable; without it, test temporal transfer and state the limits. Failure to add information is a valid, prespecified outcome and should be reported with the same rigor.

25.13.6 Journal selection

Prepare a three-journal shortlist for P1 by month 9, P2 by month 15, and P3 by month 20: a primary journal, a close-scope alternative, and a specialist option. Assess audience, length, and evidence requirements before the final draft, using:

Criterion	Weight	Assessment question
Scope fit	30%	Does the journal publish comparable scientific questions, beyond shared methods?
Audience	20%	Will critical-zone researchers understand and use the findings?

Criterion	Weight	Assessment question
Contribution type	15%	Does it accept this kind of physical, methodological, or interdisciplinary study?
Data and code policy	10%	Can its availability requirements be met under LEO data rights?
Review quality and transparency	10%	Are editorial procedures, timing, and requirements clear?
Reputation and indexing	10%	Is it recognized by the relevant community and PhD program?
Publication cost	5%	Is there a viable route without unbudgeted expenditure?

Consider impact factor alongside scope and audience. A broad journal is appropriate when the paper offers a general conclusion, external validation, and significance beyond one sensor network. A specialist venue may provide a better review of a primarily technical or local contribution.

Check the official scope, related papers from the preceding two years, article types, length limits, preprint rules, data requirements, costs, and indicators of questionable publishing practices. Record the final rationale in the project log.

25.13.7 Preparation, internal review, and submission

Each paper passes the same readiness gates:

- R1. *Concept*:** coauthors agree on the main question, audience, and claim table.
- R2. *Data*:** freeze the data version, exclusions, and training/validation split.
- R3. *Computation*:** principal results reproduce from a clean state and pass automated checks.
- R4. *Evidence*:** link every main claim to a figure or table, uncertainty, and a suitable null model.
- R5. *Internal review*:** someone who did not write the relevant code checks the logic and reproduction.
- R6. *Submission*:** satisfy journal requirements and agree on authorship, conflicts, data availability, and the cover letter.

Develop the draft from evidence: a claim–evidence–limitation table, figure captions, results, methods, discussion, and finally the introduction and abstract. This keeps the narrative within what the data support.

Before submission, perform four distinct reviews:

- scientific: question, methods, and causal wording;
- statistical: data splits, multiple testing, and uncertainty;
- reproducibility: clean-environment execution and agreement between manuscript values and outputs;
- editorial: structure, language, figures, references, and journal requirements.

25.13.8 Responding to review and rejection

Include editorial work in the two-year schedule:

1. Within two working days, enter the decision and comments into a response table.
2. Separate required corrections, scientific questions, optional analyses, and requests beyond the available data.
3. Within a week, agree which changes strengthen validation and which require an explanation of limitations.
4. For each response, identify the action, revision location, and scientific rationale.
5. Run additional analyses through the same reproducible workflow; explain any replacement of prespecified results.
6. After desk rejection, adapt the manuscript to the backup journal within ten working days.
7. After substantive rejection, address the general scientific weaknesses before selecting another journal.
8. Do not submit the same manuscript to multiple journals simultaneously.

A reviewer request does not justify an unsupported causal claim. Where the design or covariates cannot identify the requested conclusion, explain that limit and clarify the manuscript.

25.13.9 Publication milestones

Month	Required output	Completion criterion
1	Roles agreement and decision log	Every critical role has an owner
4	P1 outline and evidence table	Each planned figure addresses a question
6	P1 data and methods	Text matches dataset 1.0
9	Frozen P1 analysis	Thresholds fixed before external validation
10	Full internal P1 draft	Results, limitations, and appendices included
12	P1 submission	R1–R6 passed
13	P3 outline and null-model protocol	Minimum question fixed before inspecting final topology
15	Frozen P2 methods	Reproducible unfolding, reconstruction, and selection rules
17	Full internal P2 draft	Preservation or loss of physical dynamics documented
18	P2 submission	R1–R6 passed
20	Frozen P3 analysis	Surrogate and sensitivity tests complete
21	P3 external validation	No threshold retuning; interpretation fixed

Month	Required output	Completion criterion
22	Full internal P3 draft	Negative findings and transfer limits included
24	P3 submission and P1–P2 responses	P3 passes R1–R6; editorial actions documented

The minimum publication outcome is three complete, internally reviewed, submitted manuscripts: P1 by month 12, P2 by month 18, and P3 by month 24. Acceptance depends on external review and is not fully controllable. Manage scientific completeness, timely submission, response quality, and reproducibility. If P2 or P3 lacks a strong positive result, retain the manuscript as a rigorous assessment of applicability, incremental value, or transferability.

25.14 Risks and fallback options

Risk	Early signal	Prevention	Fallback
Incomplete metadata	Ambiguous units or depths	Check with LEO in Q1	Exclude unverified channels
Sensor drift	Persistent divergence between neighbors	Control intervals and calibration logs	Relative-change and sensitivity analyses
Too few complete events	Wide bootstrap intervals	Build the event catalog early	Continuous-scale analysis; exploratory event claims
Low coherence	Unstable phase	Quality thresholds and null models	Event lags without phase interpretation
Unidentifiable physics	Similar responses from different mechanisms	Joint amplitude, phase, and covariate analysis	Regime classification without unique attribution
Unfolding sensitivity	Conclusions change with n or q	Parameter-range analysis and bootstrap	Diagnostic rather than central role for WP5
Sparse spatial sampling	Unstable H_1 loops	Interpolation ensembles and simulations	Restrict to H_0 and graph connectivity
No independent validation	Missing ERT or chemistry	Check availability in Q1	Claims about CO ₂ dynamics only
Overambitious schedule	Delayed M3 or M4	Minimum independent scope for each paper, shared P0, continuous writing	Narrow P3 to robustness and transfer limits; retain the third manuscript

25.15 Criteria for a successful program

Assess the PhD by the quality of its evidence. By completion:

1. every observation and quality-control decision has documented provenance;

2. the physical baseline has been tested before advanced descriptors are introduced;
3. parameter fitting is separated from external validation;
4. uncertainty accounts for temporal and spatial dependence;
5. unfoldings, phases, and topological descriptors pass their own robustness checks, or their failures are documented;
6. incremental value over simple baselines is demonstrated or its absence established;
7. reversible hydrological responses are separated from residual drift as far as the data permit;
8. independent observations support the main interpretation, or it is explicitly limited to CO₂ field dynamics;
9. manuscripts, thesis, code, and permitted data form one reproducible research record.

Year one provides a self-contained study of transport physics and data analysis. Year two tests the proposed integration of Buchstaber unfoldings, phase geometry, and topology. The thesis therefore remains defensible if individual topological tests are negative, while positive findings must earn their physical and statistical interpretation.

26 Terminology for publication

A suitable method name is:

Lag-informed phase–topology analysis of distributed soil CO₂ dynamics.

To state the connection with Deymier more narrowly:

A geometric-phase-inspired framework for distributed CO₂ sensor fields.

Avoid “topological acoustic sensing,” since no acoustic waves are measured. Do not label every cluster of observations a “topological phase” without establishing a robust discrete regime or a change in an invariant.

A possible paper title, subject to the evidence for soil development, is:

Depth-Resolved Phase and Topology of CO₂ Transport during Incipient Soil Formation in a Controlled Basalt Hillslope.

A more cautious alternative is:

Geometric Phase and Topology of Distributed CO₂ Dynamics in an Evolving Basalt Landscape.

27 Conclusions

Delays between CO₂ signals in the air above the surface and at 5, 25, and 50 cm provide a physical starting point for a phase–topology description of LEO. Lags and phase gradients characterize propagation; coherent lead–lag relationships define a directed network; and completed rainfall cycles trace normalized complex states with a geometric phase. In parallel, persistent homology describes spatial domain connectivity and loop-like trajectory geometry.

Buchstaber unfoldings connect the original records to this interpretation through an intermediate geometric representation. Overlapping windows form an oriented curve whose linear approximation error can be quantified and whose optimal subspaces can be compared across regimes. Reconstruction as an L -approximation makes lag and phase preservation directly testable. The proposed LEO contribution lies in linking this established method to depth-resolved transport physics and comparisons of reversible responses and residual change.

The central scientific test is whether these descriptors distinguish reversible hydrological effects from lasting reorganization. That requires moisture- and temperature-matched dry states, parallel analysis of covariate-adjusted residuals, and validation against ERT, water chemistry, mineralogy, and biology. With these checks, a distributed CO₂ network can support a description of how the organization of an entire basalt hillslope changes over time, with the strength of the interpretation set by the evidence.

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